Machine learning
- Introduction

CdL Informatica e Innovazione Digitale

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Outline

1. General information about the course

2. What is machine learning?
   - Definitions

Course Info

- **Docente:** Valerio Freschi
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- **Ricevimento:** Martedì 11-13 oppure su appuntamento

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- **Risorse e materiale didattico:** https://blended.uniurb.it/moodle/
- **Programma e informazioni sul corso:** https://www.uniurb.it
- **Testi di studio:**
  - J. Watt, R. Borhani, A. Katsaggelos: "Machine Learning Refined. Foundations, Algorithms, and Applications", Cambridge University Press, (2020)
  - A. Lindholm, N. Wahlström, F. Lindsten, T.B. Schön: "Machine Learning - A First Course for Engineers and Scientists", Cambridge University Press, (2022).
- **Modalità d'esame** progetto e prova orale

Course Info

Programma del corso

1. Introduzione
2. Ottimizzazione matematica
3. Regressione lineare
4. Classificazione lineare binaria
5. Classificazione lineare multi-classe
6. Apprendimento non supervisionato
7. Apprendimento non lineare
8. Attività di laboratorio (esercitazioni Python)

Outline

1. General information about the course

2. What is machine learning?
   - Definitions

What is machine learning?

- For many problems it is particularly difficult to program a computer to execute a specific task by hand
  - People and object recognition
  - Understanding human speech

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- For many problems it is particularly difficult to program a computer to execute a specific task by hand
  - People and object recognition
  - Understanding human speech
- Machine learning algorithms automatically learn from data (from experience)

What is machine learning?

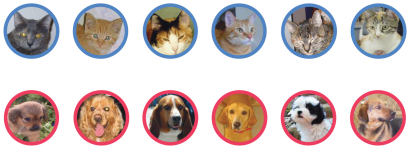
- For many problems it is particularly difficult to program a computer to execute a specific task by hand
  - People and object recognition
  - Understanding human speech
- Machine learning algorithms automatically learn from data (from experience)
- Machine learning systems are an option if:
  - coding for achieving a specific behaviour is hard (or leads to worse performance)
  - the system has to be adaptive (e.g. spam detection)

An introductory motivating example

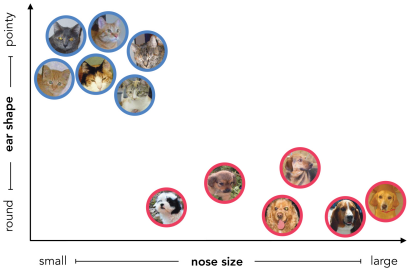
- Think of a cognitive task, e.g. distinguishing cats from dogs
- How do children learn it? By example
- They are presented many images
- A supervisor tell them if they are cats or dogs
- When new images are presented, they become able to *generalize*
- A classification task

An introductory motivating example

- Data is first collected

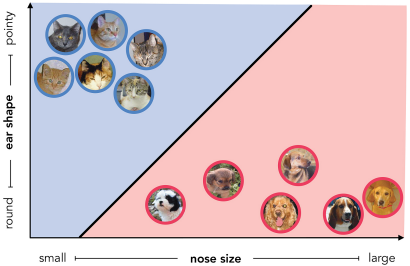


- Features are designed



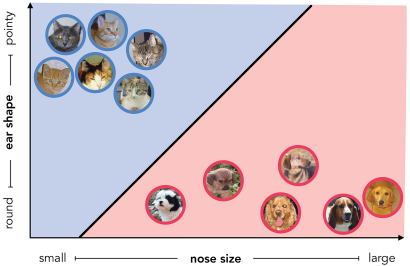
An introductory motivating example

- A model is trained....



An introductory motivating example

- A model is trained....

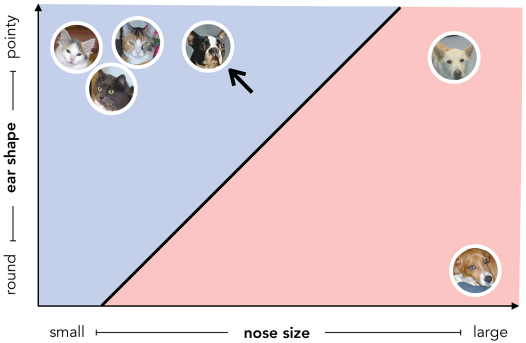


- ... and then validated



An introductory motivating example

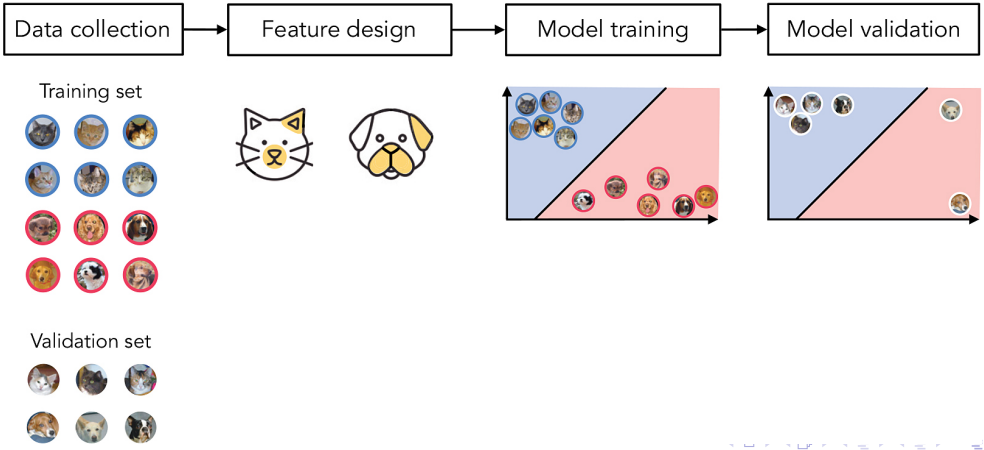
- Model validation points out the efficacy of out trained learner



- How can we improve performance?
  - collect more data
  - design more (better?) discriminating features
  - try other (e.g. nonlinear?) models

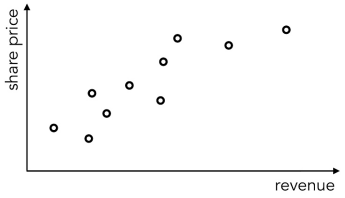
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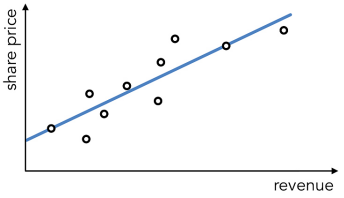
- To summarize:

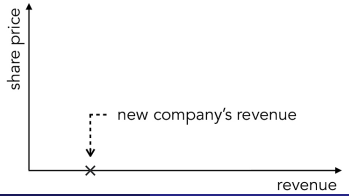


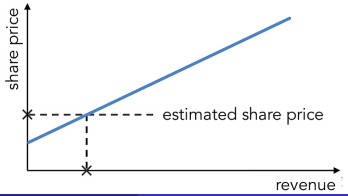
Basic taxonomy

- Supervised learning is essentially the automatic learning of computational rules with I/O relations.
- Regression, classification
- Regression:



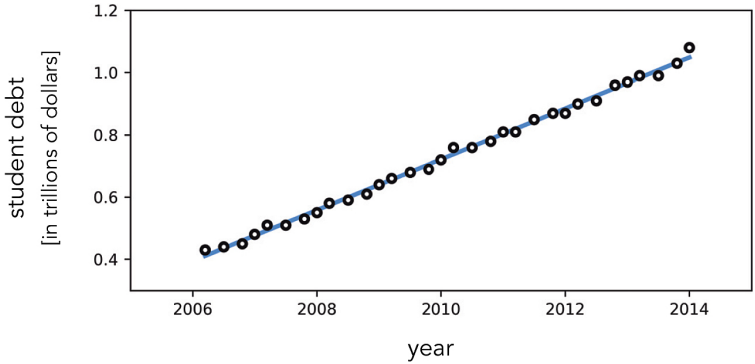






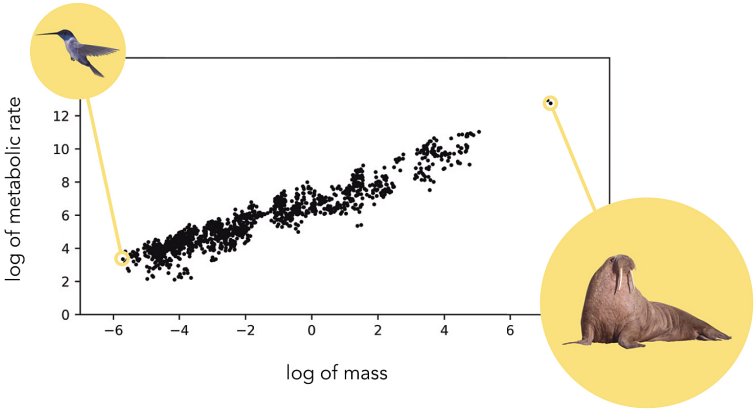
Basic taxonomy

- Regression: other examples



Basic taxonomy

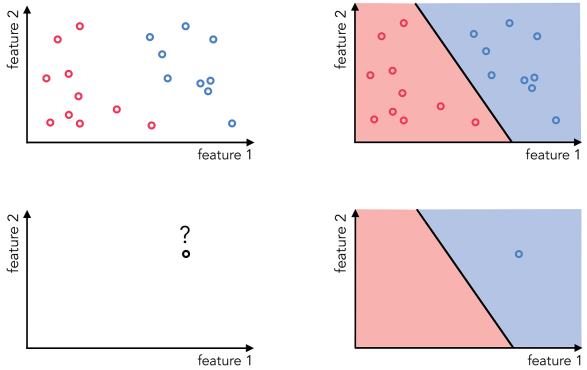
- Regression: other examples



- Many other examples in business and industry

Basic taxonomy

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- Classification:

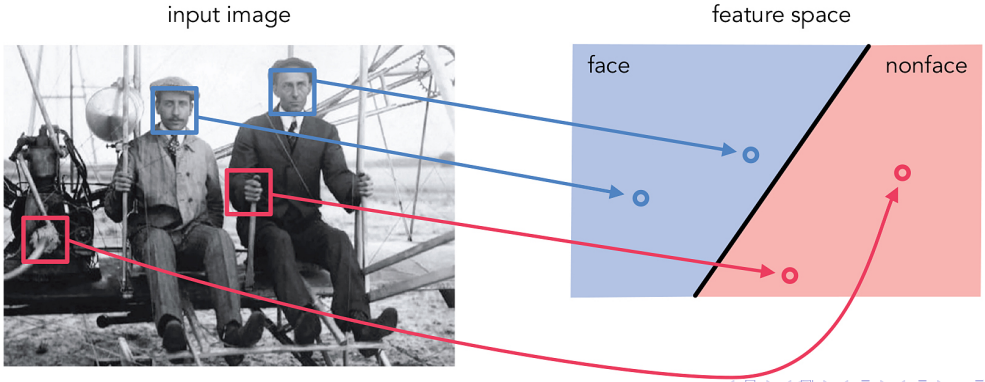


Basic taxonomy

- Classification, some examples:
  - Object detection
  - Sentiment analysis
  - Medical diagnosis
  - Spam detection
  - Finance applications (e.g. fraud detection)
  - Pattern recognition (e.g. handwritten digits)

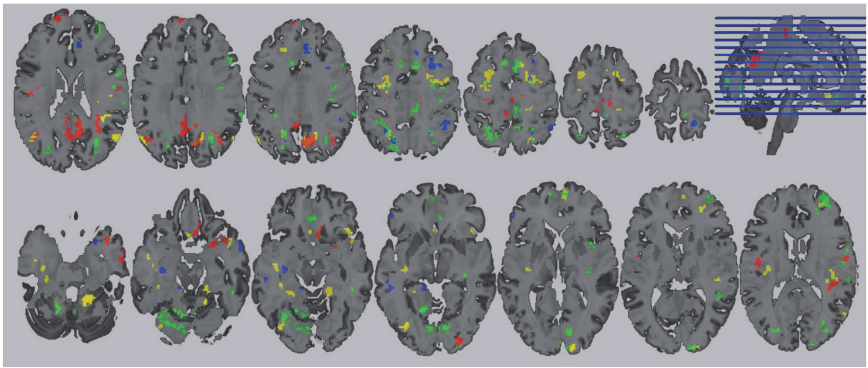
Basic taxonomy

- Supervised learning is essentially the automatic learning of computational rules with I/O relations.
- Regression, classification
- Classification: face recognition



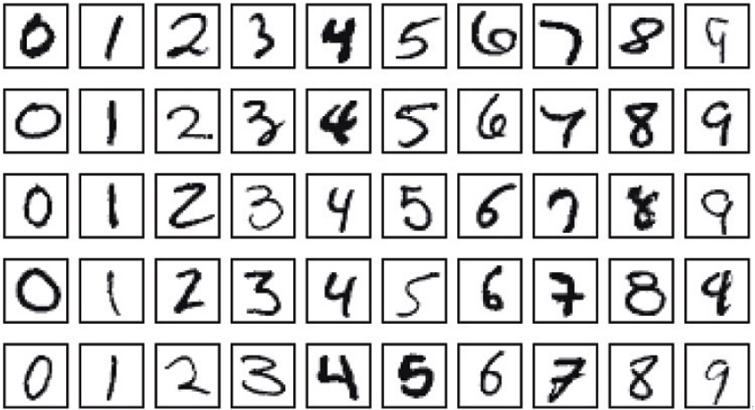
Basic taxonomy

- Supervised learning is essentially the automatic learning of computational rules with I/O relations.
- Regression, classification
- Classification: computer-aided medical diagnosis



Basic taxonomy

- Supervised learning is essentially the automatic learning of computational rules with I/O relations.
- Regression, classification
- Handwritten digits recognition



Basic taxonomy

Relies on labeled examples of correct behaviour

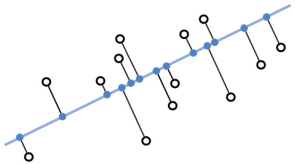
- Handwritten digit classification
  - Task: given an image of a handwritten digit, predict its class (multiclass problem)
  - Input: the image (i.e. gray-scale image)
  - Output: digit class in [0-9]
- Data example, the MNIST dataset: 70000 images of handwritten digits (a.k.a. the "fruit fly" of neural network research)
  - Training set: first 60000 images used to train the classifier (e.g. a neural network)
  - Test set: the remaining 10000 images, not available during training, used to evaluate performance
  - Performance: neural networks achieved $> 99\%$ accuracy (already 20 years ago)

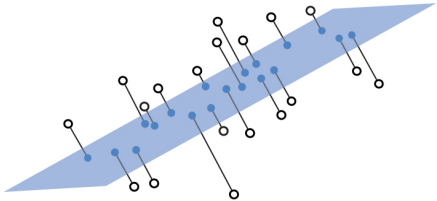
Basic taxonomy

- Face detection/recognition
- Object recognition
  - ImageNet dataset
  - Thousands categories, millions of labelled images
  - Variable conditions (viewpoint, lighting)
  - Error rate: 25.7% $\rightarrow$ 5.7% in few years
- Caption generation

Unsupervised learning: examples

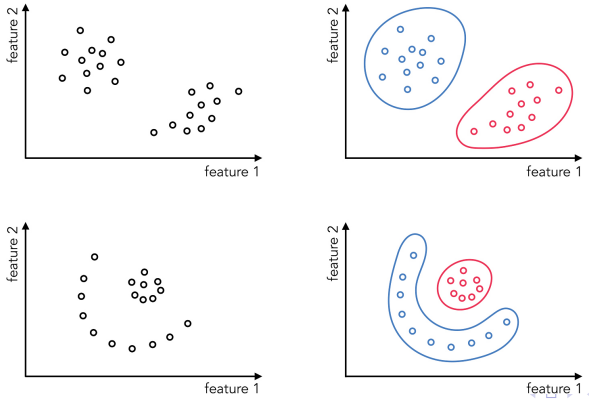
- Unsupervised learning is essentially the automatic learning of computational rules that describe input data only
- Dimensionality reduction, Clustering
- Dimensionality reduction:





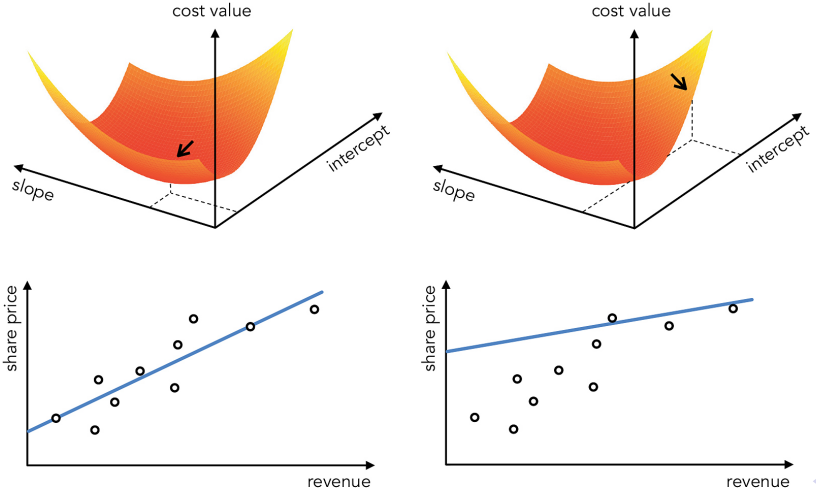
Unsupervised learning: examples

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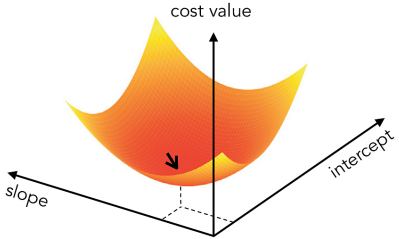
Learning algorithms rely on optimization

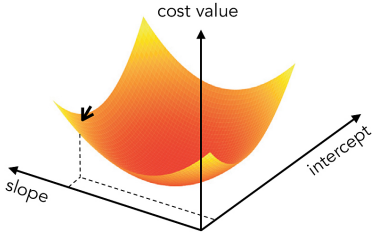
- Mathematical optimization is the framework to search for parameters of learning models
- An example with regression:

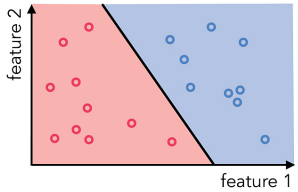


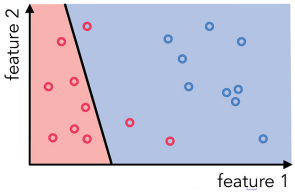
Learning algorithms rely on optimization

- Mathematical optimization is the framework to search for parameters of learning models
- An example with classification:









Recap: what is machine learning?

Machine learning

- Machine learning algorithms/systems are usually classified in:
  - **Supervised learning**: labeled examples are given that specify the correct behaviour
  - **Unsupervised learning**: no labeled examples are given, but *interesting* patterns in data have to be recognised

- **Reinforcement learning**: the system receives a reward signal and tries to learn how to maximise it